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COLLEGE OF ENGINEERING
POONJAR

GROUP 1

PROJECT

SYNAPSE

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UI/UX DESIGN DOCUMENT

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1.INTRODUCTION

1.1 Purpose

The purpose of this UI/UX Design Document is to describe the proposed user interface and user experience of the Synapse – AI-Powered Train Delay Prediction System. It explains how users interact with the application and how information is presented throughout the prediction process. The document provides an overview of the screens, navigation flow, and interface components that will be implemented in the web application.

A well-designed user interface helps users perform prediction tasks easily, while a good user experience ensures that the application is simple to understand, responsive, and efficient. The design aims to reduce user effort by requiring only essential journey details and presenting prediction results in a clear and meaningful format.

1.2 Scope

The scope of this document is limited to the proposed user interface of the Synapse web application. It describes the main screens of the system, the navigation between them, and the overall user interaction.

The document covers the Home Page, Prediction Page, and Prediction Result Page. It also includes the navigation flow and simple wireframes that represent the basic layout of each screen. The design is intended to provide users with a straightforward method of obtaining train delay predictions without unnecessary complexity.

2.USER INTERFACE DESIGN

2.1 Home Page

The Home Page serves as the entry point of the application. It introduces users to the Synapse system and briefly explains its purpose. The page is designed with a clean layout that allows users to quickly understand the functionality of the application.

The Home Page contains the project title, a short description of the system, and a Start Prediction button that directs users to the prediction page. The interface uses a simple layout with clear headings and minimal visual elements to improve readability.

The objective of this page is to provide users with an easy starting point while maintaining a professional appearance.

2.2 Prediction Page

The Prediction Page is the primary interface where users provide journey information required for delay prediction. The page contains a simple form with clearly labeled input fields.

The user enters the following information:

- Train Number
- Journey Date
- Boarding Station
- Destination Station

After entering the required details, the user clicks the Predict Delay button. The entered information is validated before being sent to the backend server for processing. If any required field is missing or contains invalid information, an appropriate validation message is displayed. The prediction page is designed to minimize user input while collecting sufficient information for the machine learning model to generate accurate predictions.

2.3 Prediction Result Page

The Prediction Result Page displays the output generated by the machine learning model. The information is organized in a simple and readable format so that passengers can quickly understand the predicted delays.

The page displays:

- Selected Train Number
- Predicted delay at each station
- Confidence score of the prediction
- Major factors influencing the prediction

The contributing factors may include weather conditions, weekend travel, historical delay patterns, or other contextual information used by the prediction model.

The page also allows users to return to the prediction page and perform another prediction if required.

3.USER EXPERIENCE DESIGN

3.1 User Flow

The user interaction within Synapse follows a simple sequence that minimizes unnecessary steps. The application guides the user from entering journey information to viewing prediction results through an intuitive workflow.



Know before the platform does.

Pick a train and a journey date. Synaps predicts the delay at every station ahead — and why.

TRAIN

12625 · Kerala Express

JOURNEY DATE

2026/08/05

BOARDING STATION

Thrissur

DESTINATION STATION

Kollam

Weather, holidays, and historical patterns for this route are gathered automatically.

[Predict delay](#)

Run a prediction to see the station-by-station outlook.

Synaps · model retrains periodically on new operational data · predictions are estimates, not guarantees.

prediction confidence

92%

probability of delay

76%

Palakkad → Ernakulam Jn

Palakkad
departure

on time

```
</div>
</div>

<div class="synaps-tl-row">
  <div class="synaps-tl-dot" style="background:#FFFFFF;border-color:#D9A441;"></div>
  <div class="synaps-tl-content">
    <div class="synaps-tl-station">Thrissur</div>
    <div class="synaps-tl-sub">stop 1</div>
  </div>
  <div class="synaps-tl-right">
    <div class="synaps-mono synaps-tl-value" style="color:#D9A441;">+14 min</div>

    <span class="synaps-pill" style="background:#FBF3E2;color:#D9A441;">
      <span class="synaps-pill-dot" style="background:#D9A441;"></span>Minor delay
    </span>
  </div>
</div>

<div class="synaps-tl-row">
  <div class="synaps-tl-dot" style="background:#FFFFFF;border-color:#C9694A;"></div>
  <div class="synaps-tl-content">
```

3.2 Navigation

The navigation structure of the application is simple and easy to understand. Users begin at the Home Page and proceed directly to the Prediction Page using the Start Prediction button.

After submitting the required journey details, users are redirected to the Prediction Result Page where the predicted delay information is displayed. Users can then return to the Prediction Page to generate another prediction if necessary.

The navigation structure reduces confusion and provides a smooth user experience by limiting the number of pages involved in the prediction process.

4. WIREFRAMES

The following wireframes illustrate the basic layout of the proposed user interface. These sketches represent the placement of interface components and are intended only for design planning.

Home Page

SYNAPSE

AI-Powered Train Delay Prediction System

Predict train delays before your journey.

[Start Prediction]

Prediction Page

Train Delay Prediction

Train Number

[_____]

Journey Date

[_____]

Boarding Station

[_____]

Destination Station

[_____]

[Predict Delay]

Prediction Result Page

Prediction Result

Train Number : 12625

<i>Station</i>	<i>Predicted Delay</i>
----------------	------------------------

<i>Thrissur</i>	<i>40 Minutes</i>
-----------------	-------------------

<i>Ernakulam</i>	<i>35 Minutes</i>
------------------	-------------------

<i>Kottayam</i>	<i>25 Minutes</i>
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Confidence : 90%

Reasons

- *Heavy Rainfall*
- *Weekend Passenger Rush*
- *Historical Delay Pattern*

[Predict Another Journey]

5.CONCLUSION

The proposed UI/UX design for the Synapse system focuses on providing a simple, user-friendly, and efficient interface for passengers seeking train delay predictions. By minimizing the amount of information that users need to enter and presenting prediction results in a structured format, the system enhances usability and improves the overall user experience.

The clean navigation flow and intuitive interface allow users to complete the prediction process with minimal effort. As the project evolves, the interface can be further enhanced by incorporating additional features such as responsive design, interactive visualizations, personalized notifications, and improved accessibility, ensuring that the application remains practical and easy to use for a wide range of users.

6.REFERENCES

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